Mapping Linked Genes

Biology 2e chapter 13 reading connection | Student worksheet / preparation

Name: _____ Class: _____ Date: _____

Before predicting

A testcross can reveal the gametes produced by a heterozygous parent. Identify the parental allele combinations before classifying offspring; recombinant means a new combination relative to those parents.

Investigation question

Do two traits assort independently, or is there evidence of linkage?

Prediction and reasoning:

Controls, independent variable, measurement and sampling plan:

How to use the virtual lab

The modeled cross is a female GL/gl x male gl/gl testcross, with complete dominance for gray body and long wings. Gray/long and dark/short are parental combinations; the other two are recombinant. Inspect the offspring on the sorting pad with the magnifier. Tally the four body/wing combinations using counters; optional marks help track counted flies. Your tallies are preserved.

Use the equipment tray to prepare the sorting pad. Select a fly to inspect it through the magnifier. Use the four manual tally counters and optional inspection marks; record both body and wing traits. Review your total before collection. Review observations, then click Collect data and continue. Complete the assigned comparison before exploring.

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Biology 2e chapter 13 reading connection | Student worksheet / observations

Label every sample virtual or classroom. Keep independent replicates separate; retain unexpected results. Units: flies

Condition / sample	Repeat	Source	Gray / long	Dark / short	Gray / short	Dark / long
Offspring sample A	1					
Offspring sample B	1					
Offspring sample C	1					

Equipment and observations

Record calibration, initial values, specimen selection, ambiguous observations or method changes. Use additional paper for larger classroom counts.

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Biology 2e chapter 13 reading connection | Student worksheet / detailed record

Sample / field: _____ Source: _____

Item / band / cell	Measurement or classification	Reasoning / uncertainty

For large counts, use tallies with a clear starting position so each specimen is counted once. For microscopy and gels, record calibration and measurement units.

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Biology 2e chapter 13 reading connection | Student worksheet / interpretation

Identify parental and recombinant totals. Calculate recombinants / total x 100.

Why can a small observed recombination percentage underestimate distance between widely separated genes?

Why is the sex of the heterozygous parent relevant in Drosophila?

Explore after the comparison

Propose one additional test. Change one condition or sample choice, collect exploration separately, and explain what the new evidence adds.

Important limits: Original stylized flies and simulated offspring under an illustrative 18% recombination probability. No actual supplier stock dataset or viability effects are represented. Two-point mapping cannot establish three-gene order.